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1 Methodology

Since our objective is to study family size, there is no direct variable in the dataset that represents this concept. Therefore, we assume that each household in the sample has the same number of elder family members, including parents and grandparents. This assumption allows us to use “Number of Children” as a proxy for family size and treat it as our response variable. Since “Number of Children” is clearly a positive integer, we first considered using Poisson regression as our modeling approach. However, since each observation has a different exposure time, we need to ensure the fairness of the model by making the count data from different observational units comparable. To achieve this, we introduce an offset variable to represent the exposure level. Specifically, we divide “Number of Children” by this offset, transforming our response variable into a ratio-based measure, ensuring that the model reasonably explains the event occurrence under a per-unit exposure framework.

Considering that Poisson regression doesn’t have a variance parameter and assumes that the mean and variance are equal, it may not perform well when applied to overdispersed data —i.e., when the variance is significantly larger than the mean. Therefore, we use the Negative Binomial model as our alternative model. In this Negative Binomial model, we maintain the same link function, offset, and variable levels. The only change is that the model assumption shifts from the number of children following a Poisson distribution to following a Negative Binomial distribution.

Upon validating the models’ applicability, we conducted a hypothesis test to evaluate the statistical significance of the models’ predictions. The hypotheses are formulated as follows:

$$H_0 : \beta_i = 0, \quad H_1 : \beta_i \neq 0$$

where β_i denotes the regression coefficient of the predictor variable under examination.

The statistical inference is based on the z-statistic, computed as:

$$z = \frac{\hat{\beta}_i}{\text{SE}(\hat{\beta}_i)} \sim N(0, 1)$$

where $\hat{\beta}_i$ represents the maximum likelihood estimate of β_i , and $\text{SE}(\hat{\beta}_i)$ is the standard error of the estimated coefficient.

To quantify the uncertainty associated with the parameter estimates, the 95% confidence interval (CI) for the exponentiated coefficient, which corresponds to the rate ratio in Poisson regression, is given by:

$$\exp(\text{CI}_{95\%}) = \left[\exp\left(\hat{\beta}_i - 1.96 \times \text{SE}(\hat{\beta}_i)\right), \exp\left(\hat{\beta}_i + 1.96 \times \text{SE}(\hat{\beta}_i)\right) \right]$$

If the confidence interval excludes 1, the predictor variable has a statistically significant effect on the response variable at the 95% confidence level in the exponentiated model summary. Conversely, if the model summary is in log form (without exponentiation), statistical significance is determined by whether the confidence interval excludes 0.

2 Result

2.1 Important Variables

The distribution of the number of children in the sample population exhibits a right-skewed shape, as shown in Figure 1. The median and mean are both 2 children, indicating that most individuals have relatively small families. However, the spread is quite large, with an interquartile range (IQR) from 1 to 3 children, meaning that the middle 50% of the population has between 1 and 3 children. The maximum value of 17 children suggests the presence of extreme outliers, which significantly extend the upper range of the distribution. The histogram in Figure 1 further confirms this right-skewed pattern, where most individuals have a small number of children, but a few have very large families.

Other key variables in the dataset also reveal important trends. The months since marriage variable (Table 1) has a median of 148 months (~12.3 years) and a wide spread from 0 to 431 months, with more recent marriages being more frequent, as seen in Figure 1. The age at marriage distribution (Figure 1) shows a peak at 22-25 years, suggesting this is the most common age range for marriage, with a slight right-skew as some individuals marry later. Finally, literacy status (Table 1) is highly skewed, with 4567 individuals being literate and only 581 illiterate, a trend clearly reflected in the bar chart in Figure 1. This suggests that literacy is widespread in the sample population, with illiteracy being a relatively minor characteristic.

Table 1: Summary of Key Variables

Number of Children	Months Since Marriage	Count by Age at Marriage	Count by Literate or Illiterate
Min: 0	Min: 0.0	At 0-15: 52	Literate: 4567
1st Qu: 1	1st Qu: 73.0	At 15-18: 452	Illiterate: 581
Median: 2	Median: 148.0	At 18-20: 910	
Mean: 2	Mean: 155.2	At 20-22: 1126	
3rd Qu: 3	3rd Qu: 230.0	At 22-25: 1468	
Max: 17	Max: 431.0	At 25-30: 923	
		At over 30: 217	

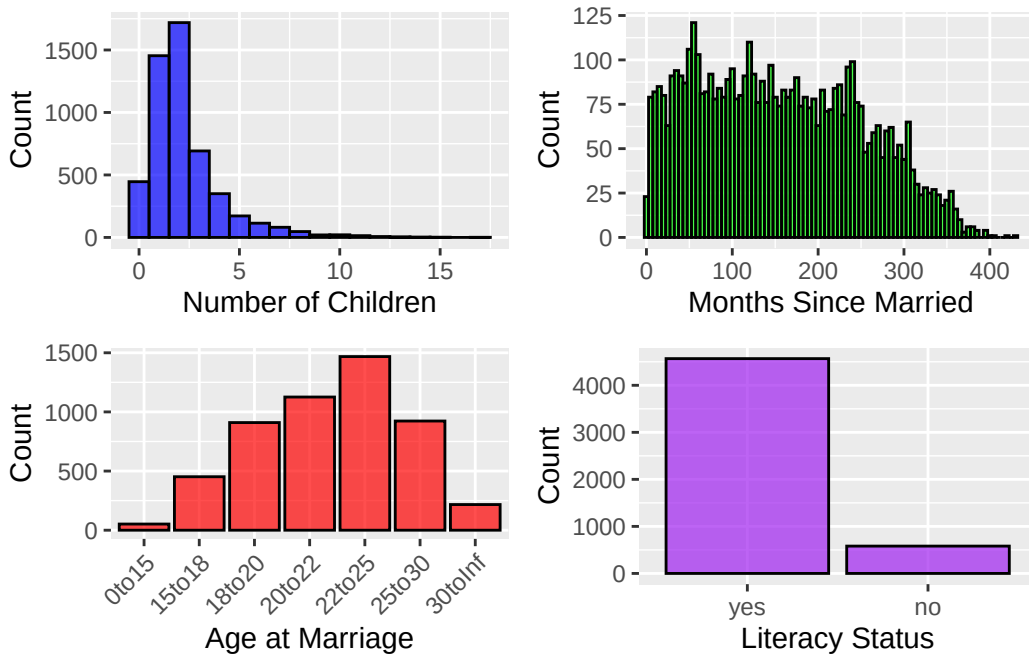


Figure 1: TBD

2.2 Model Selection

Recall our research objective: To explore the impact of literacy and age at marriage on family size (i.e., the number of children). We initially considered using Poisson regression because the number of children, as the response variable, is a non-negative integer. We included “Age at Marriage” and “Literate or Illiterate” as our predictor variables. Additionally, in Todesco’s study, we have already found that the duration of marriage has a significant effect

on the number of children (Todesco 2011). To control for the influence of different marriage durations on fertility, ensuring that the model reflects the independent effects of literacy and age at marriage, we included “Months of Marriage” as an offset to standardize the exposure period (i.e., the duration of marriage) across observations.

It is important to note that, since the link function of the Poisson distribution is log, directly taking $\log(0)$ when constructing the offset would result in negative infinity, which could affect the model’s stability and accuracy. Therefore, when constructing the offset, we ensured that the number of months of marriage for each observation was at least 1 (even if the original data contained 0s, this adjustment prevents the $\log(0)$ issue). Additionally, we divided the months by 12 to convert the time scale to years, making the model’s time representation more reasonable. By incorporating the offset, the original response variable (the number of children) is essentially transformed into a rate-based metric in the model. This means that we are now estimating the fertility rate per unit of time (e.g., per year) rather than the total number of children. With the logarithmic link function, this rate is expressed in logarithmic form, allowing the model’s results to be interpreted as the logarithm of the average number of children per year after marriage.

Furthermore, based on Table 1, we observed that the majority of our sample got married between the ages of 22 and 25. Therefore, we set the base level for the “Age at Marriage” variable to the “22 to 25 years” group. This decision allows us to interpret the effects of other age-at-marriage categories relative to the most common marriage age group in our sample.

Thus, our poisson model is:

$$\begin{aligned} \log(E[\frac{\text{Children}_i}{\text{Years}_i}]) = & \beta_0 + \beta_1 M_{0to15}(i) + \beta_2 M_{15to18}(i) + \\ & \beta_3 M_{18to20}(i) + \beta_4 M_{20to22}(i) + \beta_5 M_{25to30}(i) + \\ & \beta_6 M_{30toInf}(i) + \beta_8 \text{Literacy}_i + \\ & \epsilon_i \end{aligned}$$

Where:

$\text{Children}_i \sim \text{Poisson}(E(\text{Children}_i))$

Children_i : The total number of children born to the i th individual.

$M_{atob}(i)$: Binary indicator variable, taking the value of 1 if the i th individual entered into marriage between ages a and b , and 0 otherwise.

Literacy_i : Binary indicator variable, equal to 1 if the i th individual is illiterate, and 0 otherwise.

Year_i : The total duration of marriage (in years) for the i th individual.

$\beta_0, \beta_1, \dots, \beta_8$: Coefficients estimated in the regression model.

ϵ_i : Stochastic error term capturing unobserved heterogeneity and random fluctuations.

Since Poisson regression lacks a variance parameter and assumes that the mean and variance are equal, it may not perform well when applied to overdispersed data. Through calculation, we found that the mean number of children is 2.26, while the variance is 3.464, resulting in a variance-to-mean ratio of 1.532. This indicates that the variance is considerably larger than the mean, suggesting the presence of overdispersion. Therefore, to better model data with excess dispersion, we use Negative Binomial Regression as an alternative model:

$$\begin{aligned} \log(E[\frac{\text{Children}_i}{\text{Year}_i}]) = & \beta_0 + \beta_1 M_{0to15}(i) + \beta_2 M_{15to18}(i) + \\ & \beta_3 M_{18to20}(i) + \beta_4 M_{20to22}(i) + \beta_5 M_{25to30}(i) + \\ & \beta_6 M_{30toInf}(i) + \beta_8 \text{Literacy}_i + \\ & \epsilon_i \end{aligned}$$

Where:

$\text{Children}_i \sim \text{Negative Binomial}(E(\text{Children}_i), \theta)$

θ : Over dispersion parameter.

Children_i : The total number of children born to the i th individual.

$M_{atob}(i)$: Binary indicator variable, taking the value of 1 if the i th individual entered into marriage between ages a and b , and 0 otherwise.

Literacy_i : Binary indicator variable, equal to 1 if the i th individual is illiterate, and 0 otherwise.

Year_i : The total duration of marriage (in years) for the i th individual.

$\beta_0, \beta_1, \dots, \beta_8$: Coefficients estimated in the regression model.

ϵ_i : Stochastic error term capturing unobserved heterogeneity and random fluctuations.

2.3 Model Results

Figure 2 illustrates the predicted birth rate ratios from the Poisson model with 95% confidence intervals (CI) for different variable groups and their reference group. Since the model includes an offset for marriage duration, the estimates reflect differences in the expected birth rate (children per unit time), rather than absolute counts. Compared to individuals who married between ages 22-25 (reference group, birth rate ratio = 1.00), those who married at ages 0-15 (birth rate ratio = 1.051, 95% CI: 0.896 - 1.224, Table 2) or above 30 (birth rate ratio = 1.022, 95% CI: 0.909 - 1.145, Table 2) have slightly higher birth rates, but their confidence intervals include 1, indicating that this effect is not statistically significant. In contrast, individuals who married between 15-18 (birth rate ratio = 1.079, 95% CI: 1.009 - 1.153, Table 2) and 18-20 (birth rate ratio = 1.064, 95% CI: 1.007 - 1.123, Table 2) have significantly higher birth rates than the reference group. Their confidence intervals do not include 1, indicating a statistically

significant increase in birth rates. For individuals who married between 20-22 (birth rate ratio = 1.030, 95% CI: 0.978 - 1.084, Table 2) or 25-30 (birth rate ratio = 1.014, 95% CI: 0.958 - 1.072, Table 2), the estimated birth rate ratios remain close to 1, and their confidence intervals include 1, meaning no statistically significant difference from the reference group. For literacy, compared to literate individuals (reference group, birth rate ratio = 1.00), the estimated birth rate for illiterate individuals is 1.173 (95% CI: 1.120 - 1.228, Table 2). Since the confidence interval does not include 1, this suggests a statistically significant association between illiteracy and a higher birth rate.

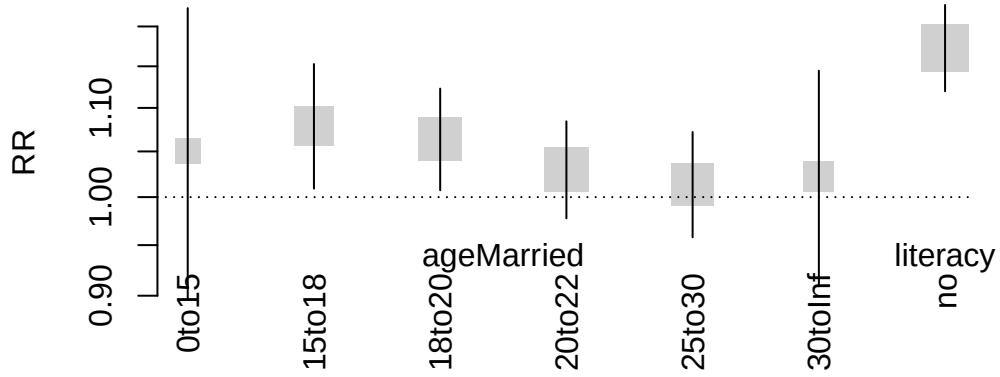


Figure 2: Relative Risk

Table 2: Summary of The Poisson Model

	Estimate	Std. Error	z value	Pr(> z)	est	2.5 %	97.5 %
(Intercept)	-1.802	0.018	-99.076	0.000	0.165	0.159	0.171
ageMarried0to15	0.050	0.080	0.624	0.533	1.051	0.896	1.224
ageMarried15to18	0.076	0.034	2.236	0.025	1.079	1.009	1.153
ageMarried18to20	0.062	0.028	2.228	0.026	1.064	1.007	1.123
ageMarried20to22	0.029	0.026	1.105	0.269	1.030	0.978	1.084
ageMarried25to30	0.013	0.029	0.468	0.640	1.014	0.958	1.072
ageMarried30toInf	0.022	0.059	0.369	0.712	1.022	0.909	1.145

Table 2: Summary of The Poission Model

	Estimate	Std. Error	z value	Pr(> z)	est	2.5 %	97.5 %
literacyno	0.159	0.024	6.770	0.000	1.173	1.120	1.228

Table 3 presents the estimation and 95% confidence intervals of the Negative Binomial model. The Negative Binomial regression model estimates the log rate of the number of children per unit of time (years). Compared to the reference group, those who married at ages 0-15 (log rate coefficient = 0.068, 95% CI: -0.109 to 0.244) or above 30 (log rate coefficient = 0.021, 95% CI: -0.102 to 0.144) have slightly higher expected log birth rates, but since their confidence intervals include 0, these effects are not statistically significant. In contrast, individuals who married between 15-18 (log rate coefficient = 0.084, 95% CI: 0.009 to 0.158) and 18-20 (log rate coefficient = 0.069, 95% CI: 0.009 to 0.129) have significantly higher log birth rates compared to the reference group. Their confidence intervals do not include 0, indicating a statistically significant increase in birth rate. For individuals who married between 20-22 (log rate coefficient = 0.035, 95% CI: -0.022 to 0.092) or 25-30 (log rate coefficient = 0.011, 95% CI: -0.051 to 0.072), the estimated log birth rate coefficients remain close to 0, and their confidence intervals include 0, indicating no statistically significant difference from the reference group. For literacy, compared to literate individuals (reference group), the estimated log rate coefficient for illiterate individuals (literacyno) is 0.148 (95% CI: 0.095 to 0.200). Since the confidence interval does not include 0, this suggests a statistically significant association between illiteracy and a higher expected birth rate. The standard deviation parameter is estimated to be 0.265, with a 95% confidence interval of 0.235 to 0.298, indicating moderate overdispersion in the Negative Binomial model. This suggests that the variance in the birth rate exceeds the mean, further supporting the choice of the Negative Binomial model over a Poisson regression.

Table 3: Summary of The Negative Binomial Model

	Estimate	97.5 %	2.5 %
(Intercept)	-1.783	-1.744	-1.822
ageMarried0to15	0.068	0.244	-0.109
ageMarried15to18	0.084	0.158	0.009
ageMarried18to20	0.069	0.129	0.009
ageMarried20to22	0.035	0.092	-0.022
ageMarried25to30	0.011	0.072	-0.051
ageMarried30toInf	0.021	0.144	-0.102
literacyno	0.148	0.200	0.095
sd	0.265	0.235	0.298

References

Todesco, Lorenzo. 2011. “A Matter of Number, Age or Marriage? Children and Marital Dissolution in Italy.” *Population Research and Policy Review* 30: 313–32.